

What is Kiro?

Kiro is an AI-powered assistant designed specifically for software development tasks. It helps developers write, understand, and transform code faster by providing contextually relevant suggestions, generating code snippets, answering technical questions, and helping troubleshoot issues. Kiro integrates directly into your development workflow through IDE plugins and CLI tools, making it accessible right where you work.

Key capabilities of Kiro include:

- **Code generation** - Create functions, classes, and entire applications based on natural language descriptions
- **Code transformation** - Refactor, optimize, and migrate existing code
- **Technical Q&A** - Get answers to programming questions and AWS service inquiries
- **Documentation assistance** - Generate comments, documentation, and test cases
- **AWS integration** - Seamless assistance with AWS services, APIs, and best practices
- **Cloud operations** - Simplify AWS resource management and infrastructure tasks

Audience

This workshop is designed for a broad technical audience including developers, architects, DevOps engineers, and technical managers who want to explore how AI can enhance their technical workflows. Whether you're writing code, designing systems, or managing technical projects, you'll discover how Kiro can help you work more efficiently. You'll learn how to leverage AI assistance to accelerate various technical tasks from coding to cloud deployment and operations.

No prior experience with AI coding assistants is required, though basic familiarity with software development concepts and AWS will be helpful.

Workshop Objectives

By the end of this workshop, you will:

- Set up Kiro in your IDE and CLI (terminal)
- Create a basic web application with Kiro's assistance
- Enhance the application with more advanced prompting techniques
- Deploy the application to AWS using either CDK or Terraform (CloudFront and S3)
- Create architecture diagrams to visualize your infrastructure
- Complete a code challenge to build a shopping cart API

Workshop Labs

- Lab 1 - Setting Up Kiro:** Choose your subscription option:
 - Option A - Pro Subscription:** Set up Kiro with Pro subscription for advanced features
 - Option B - Free Subscription:** Set up Kiro with AWS Builder ID for free access Configure Kiro in both CLI and IDE environments.
- Lab 2 - Creating a Basic and Enhanced Web Application:** Build a web application with Kiro's assistance using both CLI and IDE, generate documentation, and enhance the application with advanced prompting techniques.
- Lab 3 - Deploying with AWS:** Choose your deployment path:
 - Option A - CDK:** Deploy your application using AWS CDK with CloudFront and S3, explore created resources, and generate architecture diagrams.
 - Option B - Terraform:** Deploy your application using Terraform with CloudFront and S3, explore created resources, and generate architecture diagrams.
- Lab 4 - Simplify Cloud Operations with Kiro CLI:** Streamline AWS cloud operations by exploring CloudFormation stacks, creating VPC flow logs, and analyzing network configurations with Kiro.
- Lab 5 - MCP with Kiro:** Explore Model Context Protocol (MCP) servers to extend Kiro's capabilities:
 - MCP with Kiro CLI:** Configure and use AWS MCP servers for diagram generation, infrastructure validation, pricing analysis, and documentation access through the command line interface.
 - MCP with Kiro IDE:** Set up MCP servers in your IDE environment for enhanced development workflows.
- Lab 6 - Kiro Powers:** Explore Kiro Powers to add specialized capabilities and expert context to your Kiro agents on-demand, including hands-on experience with the Amazon Bedrock AgentCore Power.
- Lab 7 - Steering, Skills, Powers & Custom Agents:** Build a 4-layer agent architecture in Kiro:
 - Section 1: Quick Start:** Install tools, see steering in action, understand the 4 layers
 - Section 2: Core Concepts:** Build an EOL Reporting agent (Steering -> Skill -> Agent)
 - Section 3: Advanced Patterns** (Optional): Build a Terraform Reviewer (fileMatch, scripts, Powers, parallel subagents)
 - Section 4: Build Your Own** (Optional): Choose from use cases (SRE Assistant, CIS Validator, Log Analyzer, Doc Generator, IaC Expert) or bring your own
- Lab 8 - Strengthen Security with Kiro CLI:** Perform comprehensive security assessments aligned with the AWS Well-Architected Framework's security pillar, analyze CloudFormation templates for vulnerabilities, and implement security best practices with Kiro's guidance.
- Lab 9 - Financial Optimization with Kiro CLI:** Learn how to optimize costs and improve financial efficiency in your AWS environment using Kiro's recommendations for resource sizing, utilization analysis, and cost-saving opportunities.
- Lab 10 - Custom Agents:** Create specialized AI agents tailored to your specific use cases:
 - Create Your Custom Agents:** Build custom agents for specific development tasks and workflows.
 - Customize Your Custom Agents:** Fine-tune agent behavior and responses for your organization's needs.
 - Multi-Agent Collaboration:** Implement collaborative workflows between multiple specialized agents.

0. **Code Challenge:** Test your skills by building a shopping cart API for an e-commerce system following clean architecture and TDD principles using AWS services like Lambda, API Gateway, and DynamoDB, to win the prize!!! 🏆🎁

Each lab builds upon the previous one, providing a comprehensive learning experience that covers the entire development lifecycle from code creation to cloud deployment and operations.

Clean up

Clean up AWS Resources steps are provided at the end of both deployment paths (**CDK and Terraform**), **MCP sections**, and the **Code Challenge**

Benefits of Using Kiro

- **Accelerated Development** - Complete coding tasks in minutes instead of hours
- **Reduced Context Switching** - Get answers without leaving your development environment
- **Knowledge Expansion** - Learn new technologies and best practices as you work
- **Improved Code Quality** - Generate well-structured, documented code following best practices
- **AWS Expertise On Demand** - Access AWS knowledge without extensive documentation searches
- **Simplified Cloud Operations** - Streamline infrastructure management and troubleshooting

Getting Started

To begin your journey with Kiro, proceed to the [Lab 1 - Setting Up Kiro](#) where you'll configure Kiro in your development environment.

Let's start building with the power of AI assistance!

